Speciation in masked diffusion, step by step

Companion note to Part IV, §33.2 of `modelization.pdf`

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This note answers four questions about §33.2:

1. What exactly is U, what is x_U, and what does "average over x_U" mean? (§1)
2. What does the procedure look like on a short sequence? (§2)
3. Where do the profiles f come from, and why is it called "profile Bayes"? (§3, §4)
4. Can the couplings explain why the network's subfamily and clade speciation times differ from the theory? (§5)

1 The objects, one by one

- $x_0 = (x_0^1, \dots, x_0^L)$: a complete sequence of L sites. Each site holds one of q residues ($q = 20$ in the project).
- k: the **number of revealed sites**, $0 \le k \le L$.
- $U \subset \{1, \dots, L\}$: the **set of revealed sites**, with $|U| = k$. The complement M is the set of hidden (masked) sites.
- $x_U = (x_0^i)_{i \in U}$: the **revealed part**. It records *which* sites are shown and *what* residue each one holds. Example with $L = 4$ and $U = \{1, 4\}$: the partial sequence $A\,?\,?\,A$.
- $y(x)$: a **label** of a complete sequence (its family, subfamily or clade).
- $P(y \mid x_U)$: the probability that the complete sequence has label y, knowing only the revealed part.

Why x_U is random. A partial sequence is produced by two independent random choices:

1. **which sequence**: x_0 is drawn from the data distribution P;
2. **which sites**: U is drawn uniformly among the $\binom{L}{k}$ subsets of size k, *independently of* x_0. This is what masking does: it hides sites at random without looking at the residues.

The commitment averages over both choices:

$$\phi_k = \mathbb{E}_{x_0 \sim P}\, \mathbb{E}_U \Big[\sum_y P(y \mid x_{0,U})^2 \Big] = \sum_{x_0} P(x_0)\, \frac{1}{\binom{L}{k}} \sum_{U:\,|U|=k} \sum_y P(y \mid x_{0,U})^2. \tag{1}$$

The inner term $\sum_y P(y \mid x_U)^2$ is the probability that *two* clones started from this particular partial sequence end with the same label. Given x_U the two clones are independent, and both land on y with probability $P(y \mid x_U)^2$. ϕ_k is its average over all partial sequences with k revealed sites, each weighted by how often masking produces it.

With C clones. All C clones agree with probability $\sum_y P(y \mid x_U)^C$. The code does *not* use that quantity. It averages agreement over the $\binom{C}{2}$ pairs of clones, and each pair has expectation $\sum_y P(y \mid x_U)^2$. So 8 clones estimate the same ϕ_k as 2 clones, with less noise.

2 A small example you can check by hand

Take $L = 4$, two residues $\{A, B\}$, and two classes with prior 1/2 each. For this example only, assume the sites are independent within a class. The probability of residue A at each site is

site i	1	2	3	4
class 1: $f_{1,i}(A)$	0.9	0.8	0.5	0.6
class 2: $f_{2,i}(A)$	0.1	0.3	0.5	0.4

and $f_{c,i}(B) = 1 - f_{c,i}(A)$. Site 1 is very informative about the class; site 3 carries no information at all.

One sequence, all masks. Take $x_0 = AABA$ and $k = 2$. There are $\binom{4}{2} = 6$ equally likely sets U. For each one we compute the two class likelihoods, the posterior of class 1 (equal priors, so $P(1 \mid x_U) = P(x_U \mid 1)/[P(x_U \mid 1) + P(x_U \mid 2)]$), and the two-clone agreement:

U	partial sequence	$P(x_U \mid 1)$	$P(x_U \mid 2)$	$P(1 \mid x_U)$	$\sum_y P(y \mid x_U)^2$
$\{1,2\}$	$A\,A\,?\,?$	$0.9 \times 0.8 = 0.720$	$0.1 \times 0.3 = 0.030$	0.960	0.923
$\{1,3\}$	$A\,?\,B\,?$	$0.9 \times 0.5 = 0.450$	$0.1 \times 0.5 = 0.050$	0.900	0.820
$\{1,4\}$	$A\,?\,?\,A$	$0.9 \times 0.6 = 0.540$	$0.1 \times 0.4 = 0.040$	0.931	0.872
$\{2,3\}$	$?\,A\,B\,?$	$0.8 \times 0.5 = 0.400$	$0.3 \times 0.5 = 0.150$	0.727	0.603
$\{2,4\}$	$?\,A\,?\,A$	$0.8 \times 0.6 = 0.480$	$0.3 \times 0.4 = 0.120$	0.800	0.680
$\{3,4\}$	$?\,?\,B\,A$	$0.5 \times 0.6 = 0.300$	$0.5 \times 0.4 = 0.200$	0.600	0.520

Averaging the last column over the 6 subsets gives 0.736: the commitment of *this* sequence at $k = 2$. It depends strongly on the mask. It is 0.92 when site 1 is revealed, and only 0.52 when the two revealed sites are the uninformative sites 3 and 4. The full ϕ_2 of eq. (**??**) repeats this for every possible x_0 and weights each by $P(x_0)$.

What cloning does, on one row. Start from $A\,A\,?\,?$. Each clone

1. draws a class $c \sim P(c \mid x_U) = (0.96, 0.04)$;
2. fills site 3 from $f_{c,3}$ and site 4 from $f_{c,4}$.

Two clones share a class with probability $0.96^2 + 0.04^2 = 0.923$, the last column of that row. (The diffusion network does the same thing without knowing the classes. It reveals the hidden sites one at a time from its learned conditionals, and the classifier labels the completed sequence afterwards.)

How the code estimates the double average. Enumeration is impossible for $L = 60$, so the code uses Monte Carlo:

- it takes 1000 test sequences: 1000 draws of x_0;
- it gives each one a random reveal order, and U is the first k sites of that order: one draw of U per sequence, for every k;
- it completes each partial sequence 8 times, labels the clones, measures pair agreement, and averages over the 1000 sequences.

3 Where the profiles f come from

For class c, site i and residue a:

$$f_{c,i}(a) = \frac{\#\{\text{training sequences of class } c \text{ with residue } a \text{ at site } i\}}{\#\{\text{training sequences of class } c\}},$$

with a pseudo-count of 10^{-3} added before normalising, so that no entry is zero. The table $(f_{c,i})_{i=1,\dots,L}$ is the class's **profile** (a position-specific frequency table). The **profile model** pretends that sites are independent within a class:

$$P_{\text{prof}}(x \mid c) = \prod_{i=1}^{L} f_{c,i}(x^i).$$

This is an approximation. The true clade models are Potts models, whose couplings J_{ij} correlate the sites.

Why the revealed part is easy under the profile model. The likelihood of the revealed residues is the probability of the complete sequence summed over every value of the hidden sites:

$$P_{\text{prof}}(x_U \mid c) = \sum_{x_M} \prod_{i\in U} f_{c,i}(x^i) \prod_{j\in M} f_{c,j}(x^j) = \prod_{i\in U} f_{c,i}(x^i) \prod_{j\in M} \underbrace{\Big(\sum_{a} f_{c,j}(a)\Big)}_{=1} = \prod_{i\in U} f_{c,i}(x^i).$$

Each hidden site sums to 1, so hidden sites simply drop out. This is exactly what the table of §2 used.

Why the true posterior cannot be computed. For the Potts model of class c,

$$P_c(x_U) = \frac{1}{Z_c} \sum_{x_M} e^{-E_c(x_U, x_M)}.$$

The couplings link revealed and hidden sites, so the sum does not factorise. It has $q^{|M|}$ terms (up to 20^{60}), and it also needs the partition function Z_c. The profile model is the tractable substitute.

4 Profile Bayes and the class log-odds

Profile Bayes. Bayes' rule with the profile likelihood and prior π_c:

$$P_{\text{prof}}(c \mid x_U) = \frac{\pi_c \prod_{i\in U} f_{c,i}(x^i)}{\sum_{c'} \pi_{c'} \prod_{i\in U} f_{c',i}(x^i)}.$$

"Profile Bayes" means Bayes' rule applied with the profile likelihood. The theory curves in Part IV evaluate eq. (**??**) with this posterior in place of the true one, by Monte Carlo on 5000 test sequences.

The log-odds, step by step. Take two classes c, c' with equal priors. In the ratio of the two posteriors the denominator cancels:

$$\Lambda = \log \frac{P_{\text{prof}}(c \mid x_U)}{P_{\text{prof}}(c' \mid x_U)} = \log \frac{\prod_{i\in U} f_{c,i}(x^i)}{\prod_{i\in U} f_{c',i}(x^i)} = \sum_{i\in U} r_i(x^i), \qquad r_i(a) = \log \frac{f_{c,i}(a)}{f_{c',i}(a)}.$$

So the log-odds is **a sum of one term per revealed site**, and each term depends only on the two profiles at that site. That is where the dependence on f comes from.

Check on the example, row $\{1,2\}$*:* $r_1(A) = \log(0.9/0.1) = 2.197$ and $r_2(A) = \log(0.8/0.3) = 0.981$, so $\Lambda = 3.178$ and $P(1 \mid x_U) = 1/(1 + e^{-3.178}) = 0.960$, as in the table.

The mean of Λ. Let the sequence come from class c.

1. At a fixed site i the residue is drawn from $f_{c,i}$, so the average of that site's term is

$$\mathbb{E}_{a\sim f_{c,i}}\big[r_i(a)\big] = \sum_a f_{c,i}(a)\log\frac{f_{c,i}(a)}{f_{c',i}(a)} = \mathrm{KL}\big(f_{c,i}\,\|\,f_{c',i}\big).$$

2. Each site belongs to U with probability k/L, independently of its residue.
3. Summing the contributions of all sites:

$$\mu_k = \mathbb{E}\Lambda = \frac{k}{L}\sum_{i=1}^{L}\mathrm{KL}\big(f_{c,i}\,\|\,f_{c',i}\big) = \frac{k}{L}\,D.$$

D is the total information that a complete sequence carries about "c rather than c'". Revealing a fraction k/L of the sites gives, on average, that fraction of it.

Example: the KL divergence is 1.758 at site 1, 0.534 at site 2, 0 at site 3 and 0.081 at site 4, so $D = 2.37$ and $\mu_2 = \frac{2}{4}\times 2.37 = 1.19$.

From μ to a speciation time. When many weakly informative sites are revealed, Λ is approximately Gaussian with variance $2\mu_k$ (derived in §33.2 of the main document). The normalised commitment is then the universal function $\Psi(\mu) = \mathbb{E}[\tanh^2(\Lambda/2)]$, which crosses 1/2 at $\mu^\star = 1.70$. Hence

$$k_S = \frac{L\mu^\star}{D}.$$

In the project the sibling divergences are $D = 64.5,\ 23.5,\ 10.4$ nats (family, subfamily, clade). That gives $k_S = 1.6,\ 4.4,\ 9.8$ revealed sites: the family is decided first, the clade last.

5 Do the couplings explain the gap at subfamily and clade level?

The measured speciation points (k_S where the normalised commitment crosses 1/2) are:

	family	subfamily	clade
profile Bayes (theory)	2.1	4.9	8.1
network clones	2.0	5.7	16.0

Couplings affect the profile theory in two *opposite* ways.

1. **Missing information.** Clades also differ in their couplings J. A pair of revealed residues that are in contact carries class information that the profile never uses. On its own, this would make the true commitment *higher* than the profile's, and speciation earlier.
2. **Double counting.** Within a class, residues in contact are correlated, so two of them partly repeat the same evidence. The profile adds both terms r_i as if they were independent. On its own, this makes the profile posterior *over-confident*, and its speciation too early.

A test that decides which effect wins. If $Q(y \mid x_U)$ were the true posterior, then

$$\mathbb{E}\Big[\sum_y Q(y\mid x_U)^2\Big] = \mathbb{E}\big[Q(\text{true class}\mid x_U)\big].$$

The left side is the probability that two independent draws from Q agree. The right side is the probability that a draw from Q agrees with the true label, which is itself a draw from the true posterior P. When $Q = P$ the two are the same quantity. The left side is what the theory curve plots; the right side needs the true labels. If the left side is larger, Q is over-confident. The same test applied to the network is clone–clone agreement against clone–true-label agreement. On 5000 test sequences:

		profile posterior		network clones	
level	k	$\mathbb{E}\sum_y Q^2$ (plotted)	$\mathbb{E}\,Q$(true)	clone–clone	clone–true
family	5	0.873	0.873	0.871	0.879
family	20	0.997	0.996	0.999	0.999
subfamily	5	0.559	0.553	0.515	0.529
subfamily	20	0.908	0.884	0.861	0.890
clade	5	0.381	0.370	0.290	0.317
clade	10	0.605	0.562	0.430	0.479
clade	20	0.796	0.728	0.576	0.646

Reading the table.

- **Family:** the two profile columns are equal, so the profile is calibrated. Family differences (64 nats) are large compared with the correlations. Theory and network agree ($k_S \approx 2$).
- **Subfamily and clade:** the plotted profile column is too high, and increasingly so as k grows (0.796 against 0.728 at $k = 20$ for clades). *Double counting wins*, so the profile's k_S (4.9 and 8.1) is **too early**.
- **Why the error grows with k:** among k random revealed sites, the expected number of pairs in contact is $\binom{k}{2} \times \frac{179}{1770}$. That is 0.3 at $k = 3$, 2.8 at $k = 8$ and 12 at $k = 16$. For $k \leq 3$ almost no revealed pair is in contact, which is why the profile agrees with the data-only estimate there.
- **The network is under-confident:** its clone–clone agreement is *below* its clone–true agreement. This test uses no theory at all. So the network's k_S (5.7 and 16) is **too late**: its completions are spread too evenly over sibling subfamilies and clades.

Answer

Yes, couplings explain part of the disparity, but on the *theory* side. Ignoring them makes the profile theory over-confident at subfamily and clade level, so it predicts speciation too early. The network is separately under-confident, so it speciates too late. The true clade speciation lies between the two, roughly at $k \approx 9$–10 (from the calibrated column), against 8.1 for the profile theory and 16 for the network.